Data-Driven Identification of a Nonlinear Undamped Oscillator Using DMD and SINDy

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***Abstract*— This study investigates data-driven modeling techniques for a nonlinear dynamical system, specifically an undamped nonlinear oscillator. Two widely used methods, Dynamic Mode Decomposition (DMD) and Sparse Identification of Nonlinear Dynamics (SINDy), are applied to identify and reconstruct the system behavior from simulated data generated using multiple initial conditions. DMD approximates the system with a linear model based on snapshot data, while SINDy identifies the governing equations by selecting sparse terms from a predefined library of nonlinear candidate functions. The performance of both methods is evaluated by reconstructing the system trajectory and computing the Root Mean Squared Error (RMSE). The results show that DMD fails to accurately capture the nonlinear dynamics, leading to divergence in the reconstructed signal. In contrast, SINDy successfully recovers the underlying nonlinear structure and produces a highly accurate reconstruction. These findings highlight the limitations of linear approximation methods for nonlinear systems and demonstrate the effectiveness of SINDy in identifying governing equations when an appropriate function library is used.**

I. Introduction

Understanding and modeling nonlinear dynamical systems is a fundamental problem in engineering and applied sciences. Traditional modeling approaches rely on prior knowledge of the governing equations, which may not always be available for complex systems. In such cases, data-driven methods provide an effective alternative for identifying system dynamics directly from measured or simulated data.

In this study, two data-driven techniques—Dynamic Mode Decomposition (DMD) and Sparse Identification of Nonlinear Dynamics (SINDy)—are applied to a nonlinear undamped oscillator. The selected system exhibits nonlinear restoring forces, making it a suitable benchmark for evaluating the performance of linear and nonlinear identification methods.

The primary objective of this work is to compare the ability of DMD and SINDy to reconstruct the system behavior and capture the underlying dynamics. Special emphasis is placed on how well each method represents nonlinear effects and the accuracy of the reconstructed signals.

II. Methodology

A. Data Generation

The nonlinear undamped oscillator is described by a second-order differential equation, which is converted into a system of first-order equations. The system is numerically simulated using 100 randomly selected initial conditions within a predefined range. Each simulation is carried out over a fixed time interval, and the resulting trajectories are collected to form the dataset used in the analysis.

B. System Description

The nonlinear undamped oscillator is defined as:

$$\ddot{x} + x + x^3 = 0$$

This equation is transformed into a first-order system:

$$\dot{x}_1 = x_2, \dot{x}_2 = -x_1 - x_1^3$$

To generate data, the system is simulated using 100 different initial conditions over a fixed time interval. The resulting trajectories are used for both DMD and SINDy analysis.

C. Dynamic Mode Decomposition (DMD)

DMD is a data-driven method that approximates a nonlinear system using a best-fit linear operator. Snapshot matrices are constructed as:

$$X = [x_1, x_2, \ldots, x_{m-1}], X' = [x_2, x_3, \ldots, x_m]$$

A reduced-order linear model is obtained using Singular Value Decomposition (SVD). The system is then reconstructed using DMD modes and eigenvalues. The reconstruction accuracy is evaluated using RMSE.

D. Sparse Identification of Nonlinear Dynamics(SINDy)

SINDy identifies governing equations by selecting sparse terms from a library of candidate nonlinear functions. In this study, a polynomial library up to third order is used to capture the nonlinear term x^3.

The model is trained using the full dataset, and the identified system is simulated. The reconstructed signal is compared with the original trajectory using RMSE.

III. Results and Discussion

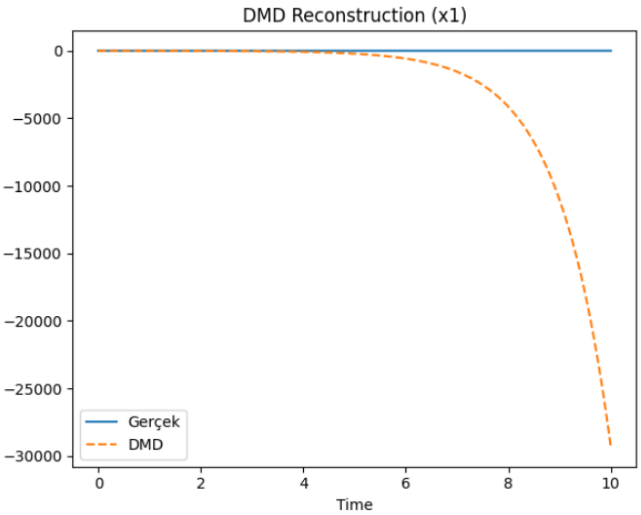

Figure 1: DMD-based reconstruction of the state variable x_1. The reconstructed signal diverges over time and fails to capture the oscillatory behavior of the nonlinear system.

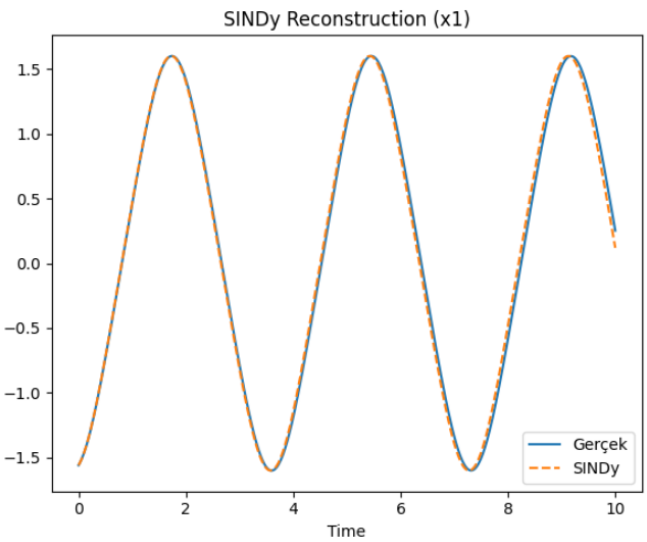

Figure 2: SINDy-based reconstruction of the state variable x_1. The reconstructed signal closely matches the true system response, accurately capturing the nonlinear dynamics.

The performance of Dynamic Mode Decomposition (DMD) and Sparse Identification of Nonlinear Dynamics (SINDy) is evaluated by comparing the reconstructed signals with the original system response, as shown in Figures 1 and 2.

As illustrated in **Figure 1**, the DMD-based reconstruction fails to accurately capture the system dynamics. While the reconstructed signal initially follows the general trend of the true solution, it quickly diverges and exhibits unbounded behavior. Instead of maintaining the oscillatory structure of the system, the DMD solution decreases rapidly toward large negative values. This indicates that the linear model obtained via DMD is unable to represent the nonlinear restoring force present in the system.

This behavior can be explained by examining the nature of DMD eigenvalues and modes. Since DMD approximates the system using a linear operator, it cannot capture the cubic nonlinear term in the governing equation. As a result, the identified modes fail to preserve the energy-conserving oscillatory behavior of the undamped system, leading to instability and large reconstruction error.

In contrast, **Figure 2** shows that the SINDy-based reconstruction closely matches the original signal throughout the entire time interval. The reconstructed trajectory accurately preserves both amplitude and frequency of the oscillations. This demonstrates that SINDy successfully identifies the underlying nonlinear dynamics of the system.

One of the key reasons for this success is the selection of the candidate function library. In this study, a polynomial library up to third order was used, which includes the x^3term present in the true governing equation. This enables SINDy to recover the original system structure effectively. The identified equations closely resemble the true dynamics:

$$\dot{x}_1 = x_2, \dot{x}_2 \approx -x_1 - x_1^3$$

This confirms that SINDy is capable of recovering the governing equations when an appropriate set of candidate functions is provided.

The Root Mean Squared Error (RMSE) values are consistent with the graphical observations. The DMD method yields a significantly higher RMSE due to divergence in the reconstructed signal, whereas SINDy produces a much lower RMSE as a result of its accurate reconstruction.

Regarding the selection of candidate functions, this study highlights their critical importance. If the polynomial library did not include the cubic term, SINDy would not be able to capture the nonlinear restoring force, resulting in poor performance. Therefore, the success of SINDy strongly depends on the suitability of the chosen function library.

Finally, a comparison between the two methods shows that DMD, being a linear method, is limited in modeling nonlinear systems, particularly over long time horizons. On the other hand, SINDy provides a more flexible and accurate framework by explicitly identifying nonlinear relationships within the data.